

Jinsong Liu

Contact Information

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Academic Background

Cornell University

Postdoctoral Associate, Weill Cornell Medicine
Research direction: AI for Healthcare

Jun. 2025 – Current

Cornell University

Visiting PhD Student, SC Johnson College of Business
Research direction: LLM Agent for Decision Making

Jul. 2024 – May. 2025

Shanghai University of Finance and Economics

Ph.D. in Management Science and Engineering

Sep. 2020 – May. 2025

Thesis: *Efficient and Adaptive Reinforcement Learning: Explorations and Improvements in Value Iteration, Policy Optimization, and Reward Preference Modeling*

Tianjin University

B.E. in Electronic Engineering
B.S. in Computer Science (Double Degree)

Aug. 2016 – Jun. 2020

Professional Experience

Pokee AI

Algorithm Research Scientist (Advisor)
Remote

Jan. 2025 – May. 2025

- Designed and implemented state-of-the-art reinforcement learning algorithms for tool-use agents, advancing capabilities in autonomous decision-making and multi-step reasoning
- Co-architected DeepResearch agents achieving SOTA performance among similar-scale open-source models through novel RL techniques and reward modeling strategies
- Open-sourced implementation (github.com/Pokee-AI/PokeeResearchOSS) garnered **1,500+ stars**, demonstrating significant impact on the AI research community

Cardinal Operations (Shanshu)

Algorithm Research Intern
Shanghai, China

Apr. 2023 – Apr. 2025

- Architected production-grade mathematical optimization solvers (LP, QP, SOCP, SDP) from scratch, combining numerical analysis expertise with efficient algorithmic design
- Pioneered deep reinforcement learning integration for optimization problems, bridging machine learning and operations research at scale

- Led development of industry’s first GPU-accelerated LP solver in C/CUDA (github.com/COPT-Public/cuPDLP-C), attracting **200+ stars** and achieving significant speedup over CPU baselines on large-scale instances
- Spearheaded CUDA optimization strategies for parallel algorithm deployment, handling problems with millions of variables and constraints

JD.com

Algorithm Research Intern
Beijing, China

Dec. 2021 – Aug. 2022

- Developed multi-agent coordination algorithms for automated warehouse systems, optimizing real-time dispatching of 100+ AGVs under dynamic constraints
- Deployed learning-based scheduling framework reducing task incompleteness rate by 90%, directly impacting production operations

Publications

†: equal contribution, ($\alpha\beta$): alphabetical order, *: corresponding author.

Working Papers & Preprints

- WP1. **Liu, J.**, Jiang, Y., Krishnan, R., Padman, R., Zhang, Y., & Bian, J. (2026). Closing Reasoning Gaps in Clinical Agents with Differential Reasoning Learning. arXiv preprint arXiv:2602.09945.
- WP2. **Liu, J.**, Brown, K., Ancker, J. S., RoyChoudhury, A., Moliere, N., Wright, A., Rosen, A., Han, J. H., Steel, P. A. D., Zhang, Y. (2025). Multi-Site Validation of Large Language Models to predict Emergency Department Return Visit With Admission. *In preparation for Nature Communications*.
- WP3. Wan, Y., Wang, J., Li, L., **Liu, J.**, Zhu, R., & Zhu, Z. (2025). PokeeResearch: Effective Deep Research via Reinforcement Learning from AI Feedback and Robust Reasoning Scaffold. arXiv preprint arXiv:2510.15862.
- WP4. Liang, K., Zhang, C., **Liu, J.***, Ge, D., & Wang, Z. (2025). A Column-Generation-Based Framework for Dynamic Network Optimization with Service Customization. Available at SSRN 5391463.
- WP5. **Liu, J.**, Ge, D., & Zhu, R. (2024). Reward learning from preference with ties. arXiv preprint arXiv:2410.05328.
- WP6. Qin, H., Zhao, X., **Liu, J.***, Ge, D., & Zhu, R. (2024). Dispatching Automated Guided Vehicles Using Efficient Data-Driven Optimization. Available at SSRN 4959037.
- WP7. Lu, H., Yang, J., Hu, H., Huangfu, Q., **Liu, J.**, Liu, T., ... & Ge, D. (2023). cupdlp-c: A strengthened implementation of cupdlp for linear programming by c language. arXiv preprint arXiv:2312.14832.
- WP8. **Liu, J.†**, Xie, C.†, Deng, Q., Ge, D., & Ye, Y. (2023). Stochastic Dimension-reduced Second-order Methods for Policy Optimization. arXiv preprint arXiv:2301.12174.

Journal Publications

- J1. ($\alpha\beta$) Deng, Q., Feng, Q., Gao, W., Ge, D., Jiang, B., Jiang, Y., ... & Zhang, C. (2025). An enhanced alternating direction method of multipliers-based interior point method for linear and conic optimization. *INFORMS Journal on Computing*, 37(2), 338-359.
- J2. ($\alpha\beta$) Ge, D., **Liu, J.**, Liu, T., Tan, J., & Ye, Y. (2024). Algorithm 1053: SOLNP+: a derivative-free solver for constrained nonlinear optimization. *ACM Transactions on Mathematical Software*, 50(4), 1-24.

Conference Publications

- C1. **Liu, J.**, Xie, C., Deng, Q., Ge, D., & Ye, Y. (2024, March). Sketched newton value iteration for large-scale Markov decision processes. In *Proceedings of the AAAI Conference*

on Artificial Intelligence (Vol. 38, No. 12, pp. 13936-13944).

Open-Source Software

- **PokeeResearch-7B Agent:** Pokee's state-of-the-art 7B DeepResearch Agent that leverages web search and content reading capabilities to answer complex questions using the most up-to-date information available online.
Repository: github.com/Pokee-AI/PokeeResearchOSS
- **cuPDLP-C: A GPU-accelerated Primal-Dual LP Solver:** Industry's first GPU-accelerated LP solver implemented in C/CUDA, achieving significant speedup over CPU baselines on large-scale instances.
Repository: github.com/COPT-Public/cuPDLP-C
- **ABIP:** An implementation of an ADMM-based interior-point method for linear and conic optimization problems.
Repository: github.com/leavesgrp/ABIP

Conference Talks & Workshops

- *Multi-Site Validation of Large Language Models to predict Emergency Department Return Visit With Admission*
AMIA Annual Symposium, Atlanta, USA, November 2025.
- *LLMs for Choice Prediction: A Generalized Direct Preference Optimization Approach*
INFORMS Annual Meeting, Atlanta, USA, October 2025.
- *LLM-Based Reasoning for Return Visit Admissions in the Emergency Department*
INFORMS Annual Meeting, Atlanta, USA, October 2025.

Teaching Experience

Guest Lecturer, Operations Research for Healthcare Mar. 2026

Institution: Weill Cornell Medicine

- Introduced foundational concepts of operations research (OR), including optimization, simulation, and decision analysis, and their applications in healthcare to graduate students.
- Introduced basis of linear programming, integer programming, and how to model healthcare problems using these techniques.
- Specific case studies demonstrating OR's potential to enhance clinical workflows, resource allocation, and patient outcomes.

Guest Lecturer, AI for Healthcare Nov. 2025

Institution: Weill Cornell Medicine

- Introduced foundational concepts of large language models (LLMs) and their applications in healthcare to graduate students.
- Introduced data structures for general healthcare data and methods for preprocessing and managing such data for LLM training and fine-tuning.
- Specific case studies demonstrating LLMs' potential to enhance clinical workflows and patient outcomes.

Skills

Programming: Python, CUDA, Julia, C/C++, Matlab, SQL

Optimization/Modeling: Gurobi, COPT, MOSEK, CLP, Cplex